



First Semester 2025-26

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : SAN G511
Course Title : Sanitation Technology
Instructor-in-charge : ANUPAM SINGHAL

1. Scope and Objective:

This course aims to give the students a review of the fundamentals as well as the latest technological developments applied in the field of sanitation. Urban Drainage and Sewerage, Carbon, nitrogen and phosphorus removal & recovery; sludge treatment, Site evaluation; toilets; onsite sanitation systems; emptying and transport; established and transferring technologies for dewatering, stabilization, pathogen inactivation and nutrient management, urban low cost drainage, Innovation processes, Intro & Exercise technology selection tool / decision support system; discuss outcomes.

2. Course learning outcomes:

- a. **Sewage Treatment**
- b. **Anaerobic Treatment**
- c. **Sanitation technology for waterless system**

3. Text Books and references:

TB1. Biological Wastewater Treatment, Principles, Modelling and Design, IWA Publishing, Guanghao Chen, Mark C.M. van Loosdrecht, George A. Ekama, Damir Brdjanovic, 2020.

RB1. Faecal Sludge Management Linda Strande, Damir Brdjanovic IWA publishing, 2014.

RB2. Sewage Disposal and Air Pollution Engineering, S. K. Garg, Khanna Publishing, 2012.

4. Method of conduct of the course

Black Board, Power Point Presentation, Video Clips and Field visits

5. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Week no.	Module	Lect. no.	Reference	Learning Outcome
1	Sanitation systems, products, Functional groups, Technologies	1	Chap. 1 TB	Introduction to Sanitation Technology
1	Physico-chemical characteristics of wastewater. Biological characteristics, wastewater flows, wastewater and Biomass Fractionation	2-3	Chapter 3 T B, Lecture Notes	Students will get knowledge of Wastewater Characterization
2	Sludge retention time, hydraulic retention time, organic matter degradation, design of wastewater treatment plants	4-5	Chapter 4 T B, Lecture Notes	Understanding of Organic Matter removal



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3	Nitrification and denitrification– microbial fundamentals and application. Biological phosphate removal	6-7	Chapter 5 -6, T B, Lecture Notes	Understanding of Nitrogen and Phosphorus removal
3	Indicator organisms - Viruses, Bacteria, Protozoa, Helminthes, Removal in wastewater treatment systems, Disinfection	8-9	Chapter 8, T B, Lecture Notes	Pathogen removal from Wastewater
4	Single Pit System , Waterless Pit System without Sludge Production, Pour Flush Pit System without Sludge Production, Waterless System with Urine Diversion, Biogas System, Blackwater Treatment System with Infiltration, Blackwater Treatment System with Effluent Transport, Blackwater Transport to (Semi-) Centralized	10-13	Lecture Notes	Description of Waterless Disposal System

	Treatment system, Sewerage System with Urine Diversion			
5	Dry Toilet, Urine- Diverting Dry Toilet (UDDT), Urinal, Pour Flush Toilet, Cistern Flush Toilet, Urine- Diverting Flush Toilet (UDFT)	14-15	Lecture Notes	User Interface
5-6	Urine Storage Tank/Container, Single Pit, Single Ventilated Improved Pit (VIP), Double Ventilated Improved Pit (VIP), Fossa Alterna, Twin Pits for Pour Flush, Dehydration Vaults, Composting Chamber, Septic Tank, Anaerobic Baffled Reactor (ABR), Anaerobic Filter, Biogas Reactor	16-19	Lecture Notes	Exposure to Collection/Storage / Treatment of sludge and anaerobic system



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6-7	Jerrycan/Tank, Human-Powered Emptying and Transport, Motorized Emptying and Transport, Simplified Sewer, Solids-Free Sewer, Conventional Gravity Sewer, Transfer Station (Underground Holding Tank)	20-21	Lecture Notes	Conveyance of Wastewater
8	Settler, Imhoff Tank, Anaerobic Baffled Reactor (ABR), Anaerobic Filter, Waste Stabilization Ponds (WSP), Aerated Pond, Free-Water Surface Constructed Wetland, Horizontal Subsurface Flow Constructed Wetland, Vertical Flow Constructed Wetland, Trickling Filter, Upflow Anaerobic Sludge Blanket Reactor (UASB), Activated Sludge, Sedimentation/Thickening Ponds, Unplanted Drying Beds, Planted Drying Beds, Co-Composting, Biogas Reactor,	22-25	Lecture Notes	Exposure to Semi Centralized treatment

	POST Tertiary Filtration and Disinfection.			
9	Fill and Cover/Arborloo, Application of Stored Urine, Application of Dehydrated Faeces, Application of Pit Humus and Compost, Application of Sludge, Irrigation, Soak Pit, Leach Field, Fish Pond, Floating Plant Pond, Water Disposal/Ground water Recharge, Surface Disposal and Storage, Biogas Combustion	26-28	Lecture Notes	Use and or disposal of Wastewater
10	Environmental Benefits, anaerobic degradation of organic polymers, impacts of alternate electron acceptors, immobilization and sludge granulation, UASB reactor, process kinetics	29-30	Chap. 16 TB, Lecture Notes	Application to Anaerobic wastewater treatment



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11-12	A. Understand the importance of faecal sludge management, and the lacks and weaknesses of the faecal sludge service chain B. Understand the fundamentals of faecal sludge management C. Integrate the information necessary to analyse and select among technologies D. Know how to design well-functioning treatment technologies E. Analyse the management and planning approaches that lead to sustainable solutions Know research and innovations in terms of technology	31-35	RB 1, Lectures Notes	Faecal Sludge Management
13-14	A. Discern the different methods of sanitation planning at the city and neighbourhood levels and understand why an enabling environment is essential B. Link the systematic approach of urban sanitation to different functional	35- 40	RB 1, Lectures Notes	Planning & Design of Sanitation Systems and Technologies

	groups C. Evaluate the pros and cons of different treatment methods D. Discern sanitation solutions in their contexts, as well as different treatment options Analyse different diagnostic and analytical tools concerning institutional and financial arrangements,behaviour changes, security planning in terms of sanitary measures and the flow of faecal matter			
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6. Evaluation Scheme:

(Given below are suggestions for possible kinds of evaluation components.
Indicative only.)

EC No.	Evaluation component	Duration	Date and Time	Weightage (%)	Nature of Component
1	Mid Sem test	90 min	7/10 AN	30	Open Book
2	Quiz	30 min	Once per month	15(Best 4 out of 5)	Open/Closed Book
3	Seminar / take home assignments	-	Will be announced	20	Open Book
4	Comprehensive Examination	3 Hours	10/12 AN	35	Closed Book



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7. Chamber consultation hours: Monday 4-5 pm in room no. 2129 or through Google Meet.

8. Makeup Policy:

- a. Make-up will be granted only on genuine reasons. However, prior permission is must.
- b. For medical cases, a certificate from the concerned physician of the Medical Centre must be produced

Anupam Singhal

I/C SAN G511 (Sanitation Technology)